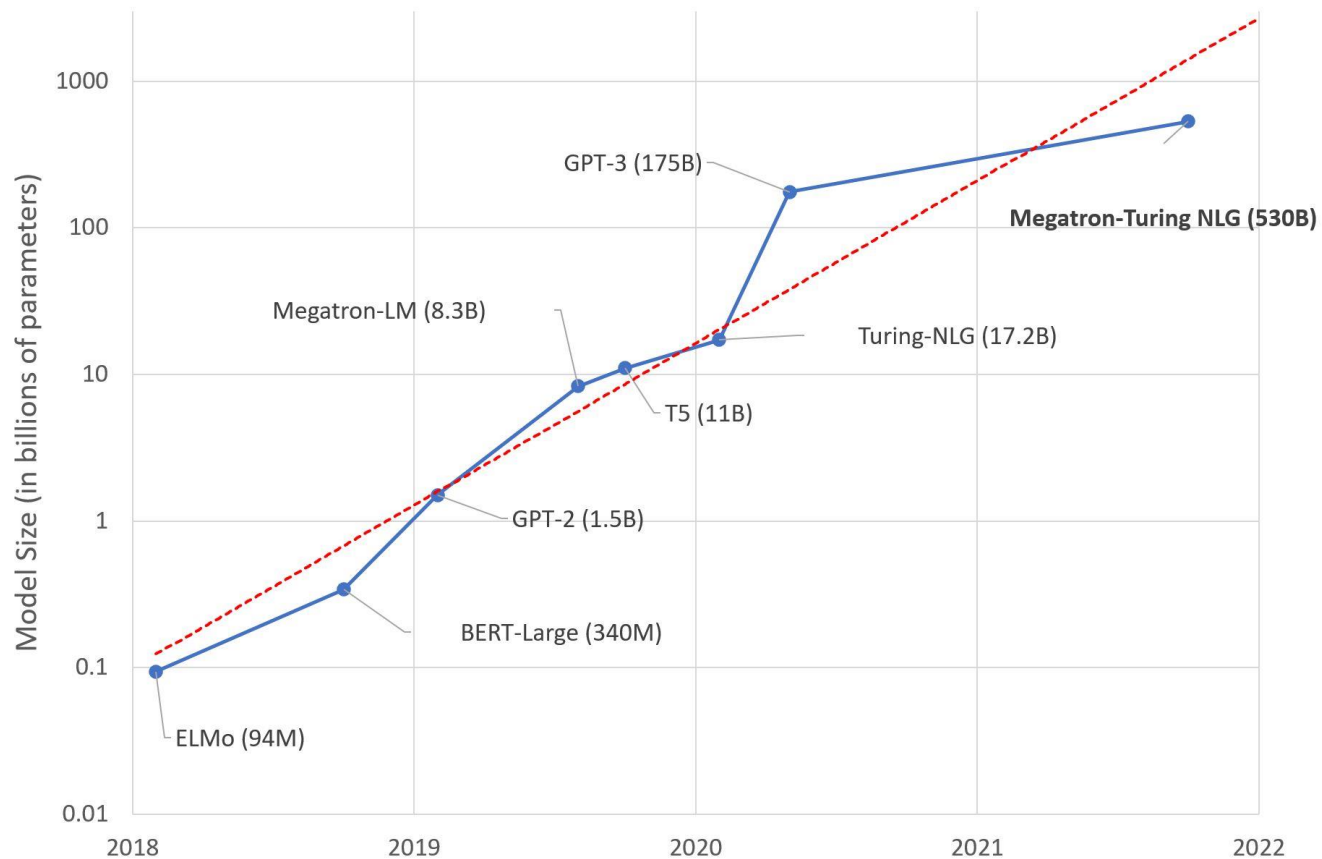


Lecture 9 – Linear Regression, and ANOVA



Today's Learning Outcomes

1. Be able to explain in graphical form how simple linear regressions operate
2. Know the differences in capabilities/limits of linear regression, multiple linear regression, general linear model, and logistic regression
3. Know how to assess explanatory variance of regression models

Comparison *versus* Regression

Assuming you work for a groundwater consulting firm, some customer in California wanted to know the groundwater flow rate at site A and site B,

- Based on what we have learned so far, if we have a series of measures for each site, we can perform a T-test, permutation resampling or other hypothesis testing techniques to examine whether a significant difference exist between sites A and B
- However, we won't be able to tell anything what might be the cause of that difference

Comparison *versus* Regression

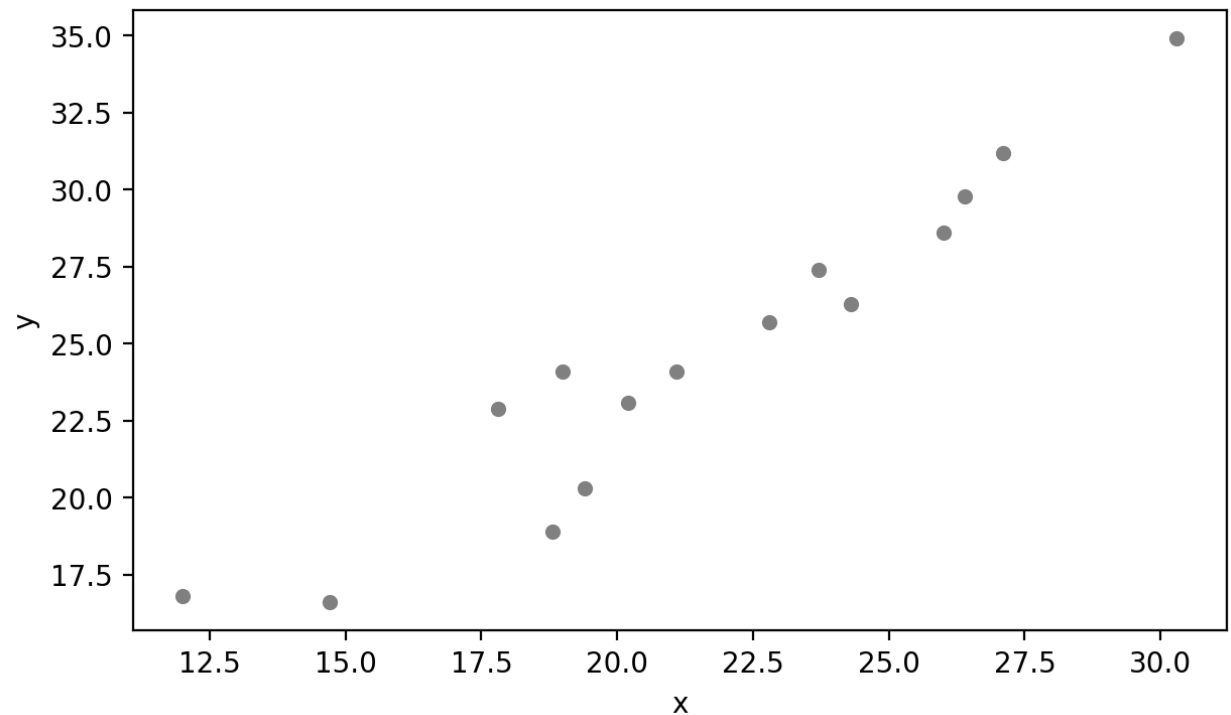
- If we also have data from each of the sites of possible explanatory variables (soil porosity, conductivity, precipitation, etc.) we can look for relationships between the outcome of interest (groundwater flow rate) and our observations
- We can estimate the character of these relationships using a regression model
 - We will stick with linear models today

Example Data

```
x = [22.8,12.0,19.4,27.1,19.0,21.1,26.4,17.8,23.7,20.2,30.3,18.8,14.7,26.0,24.3]
y = [25.7,16.8,20.3,31.2,24.1,24.1,29.8,22.9,27.4,23.1,34.9,18.9,16.6,28.6,26.3]
plt.figure(figsize=[7,4],dpi=200)
plt.scatter(x,y,s=20,facecolor='gray')
plt.xlabel('x')
plt.ylabel('y')
```

There appears to be
a positive linear
relationship
between x & y

As x increases by 1,
y increases by $A \cdot x$
where A is some
scaling factor

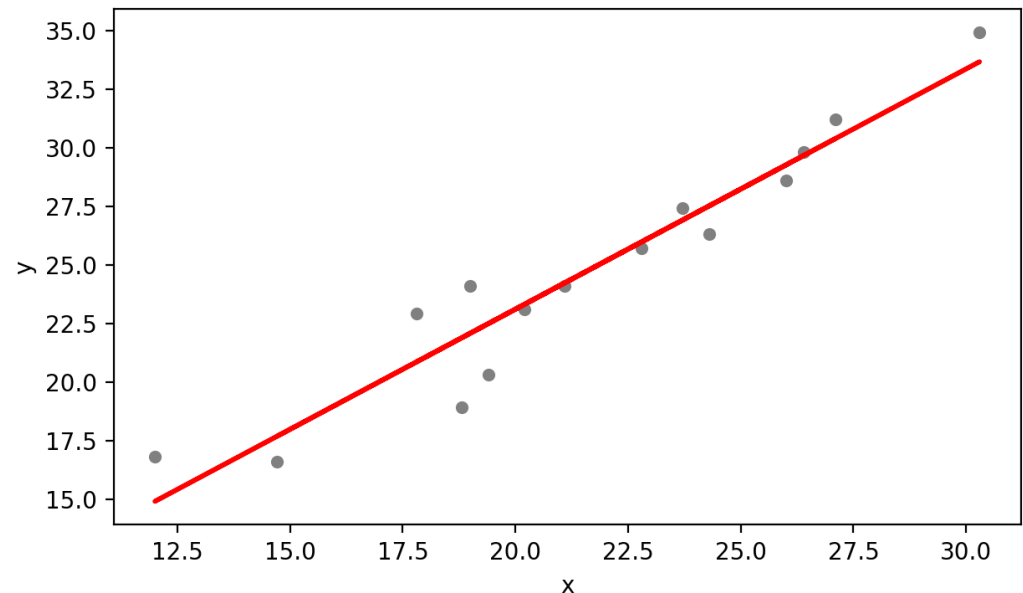


Linear Regression

- A linear regression is the estimated mathematical formula to describe how y varies as x varies
- Simplest form is of this equation is...
- is slope (how much a change in x affects the predicted value of y)
- β_0 is intercept, a constant
- ϵ is error terms (aka noise)

Linear Regression

- We can eyeball a linear regression here pretty well
 - The line here has the form $y_{predict} = (\beta_0 + \beta_1 \cdot x)$
- The e of the previous slide accounts for the difference between the predicted and observed y-values (the distance between gray dots and red lines on right plot)



Linear Regression

Python can form a much more precise line using

```
from scipy.stats import linregress  
lr = linregress(x, y)
```

$y_{predict} = (\beta_0 + \beta_1 \cdot x)$
Blue box is estimate of β_0
Red box is estimate of β_1

#Output

```
LinregressResult(slope=np.float64(1.02545287098266),  
intercept=np.float64(2.590896730000747),  
rvalue=np.float64(0.9587532973687167),  
pvalue=np.float64(1.779519609895298e-08),  
stderr=np.float64(0.08431826210956965),  
intercept_stderr=np.float64(1.8624553457473432))
```

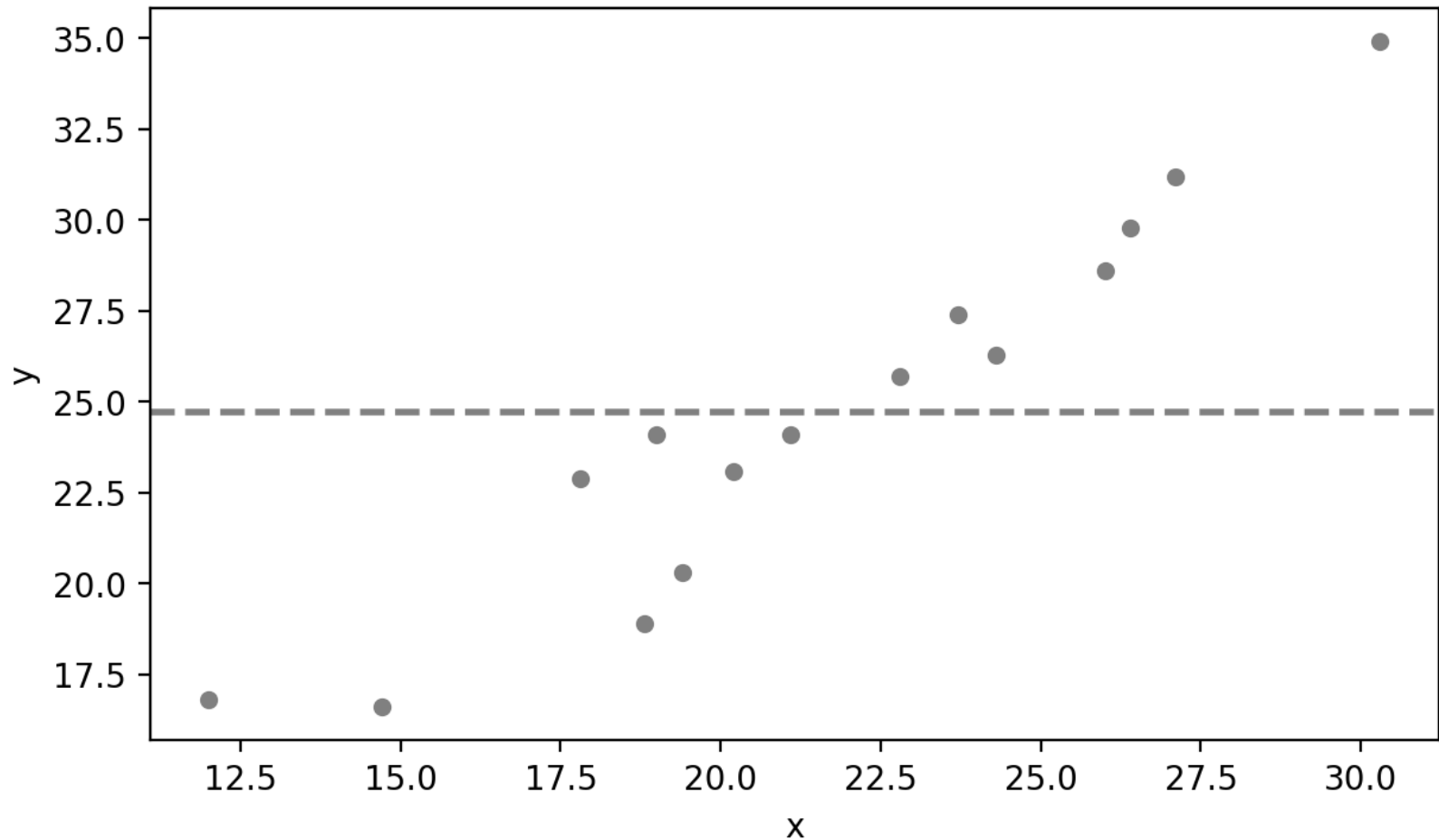

Linear Regression

- Estimating the β_0 and β_1 parameters is done by finding the line which minimizes the difference between the predicted y-values ($\mathbf{y}_{predict}$) and the observed y-values ($\mathbf{y}$), i.e., ϵ (error term)
- We can decompose the variance in our model into two parts
 - Variance explained by the model
 - Variance not explained by the model (aka error, residuals)

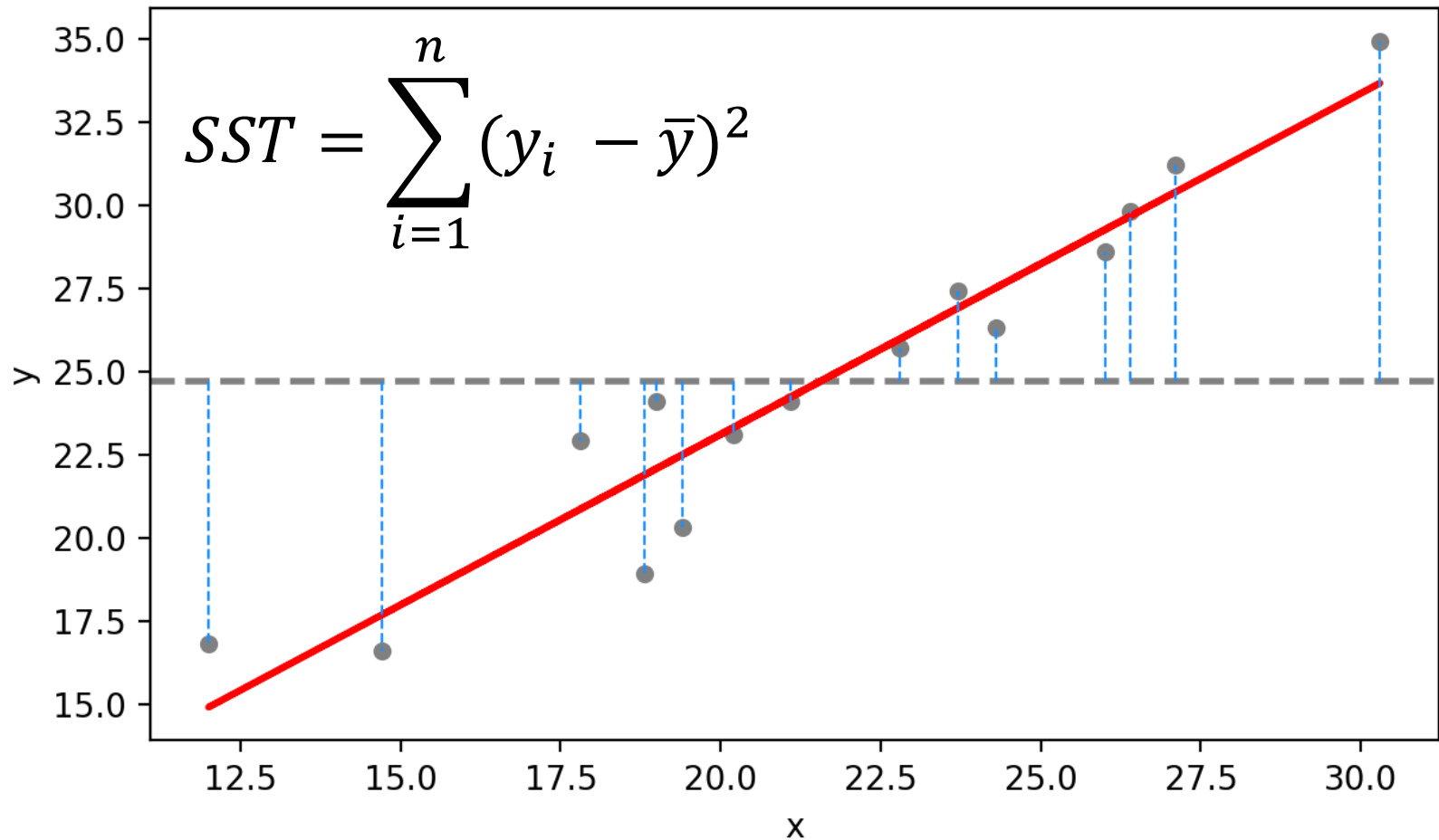
Linear Regression

- Evaluating the variance explained by the model requires we have some baseline to compare to
 - i.e. a null hypothesis
- In linear regression the null is no relationship between x and y which is where $\beta_1 = 0$
 - $y = (\beta_0 + 0 \cdot x) \rightarrow y = \beta_0$ (horizontal line)
- Most common β_0 value to use is $\text{mean}(\mathbf{y})$

Null Regression model $y = \text{mean}(y)$

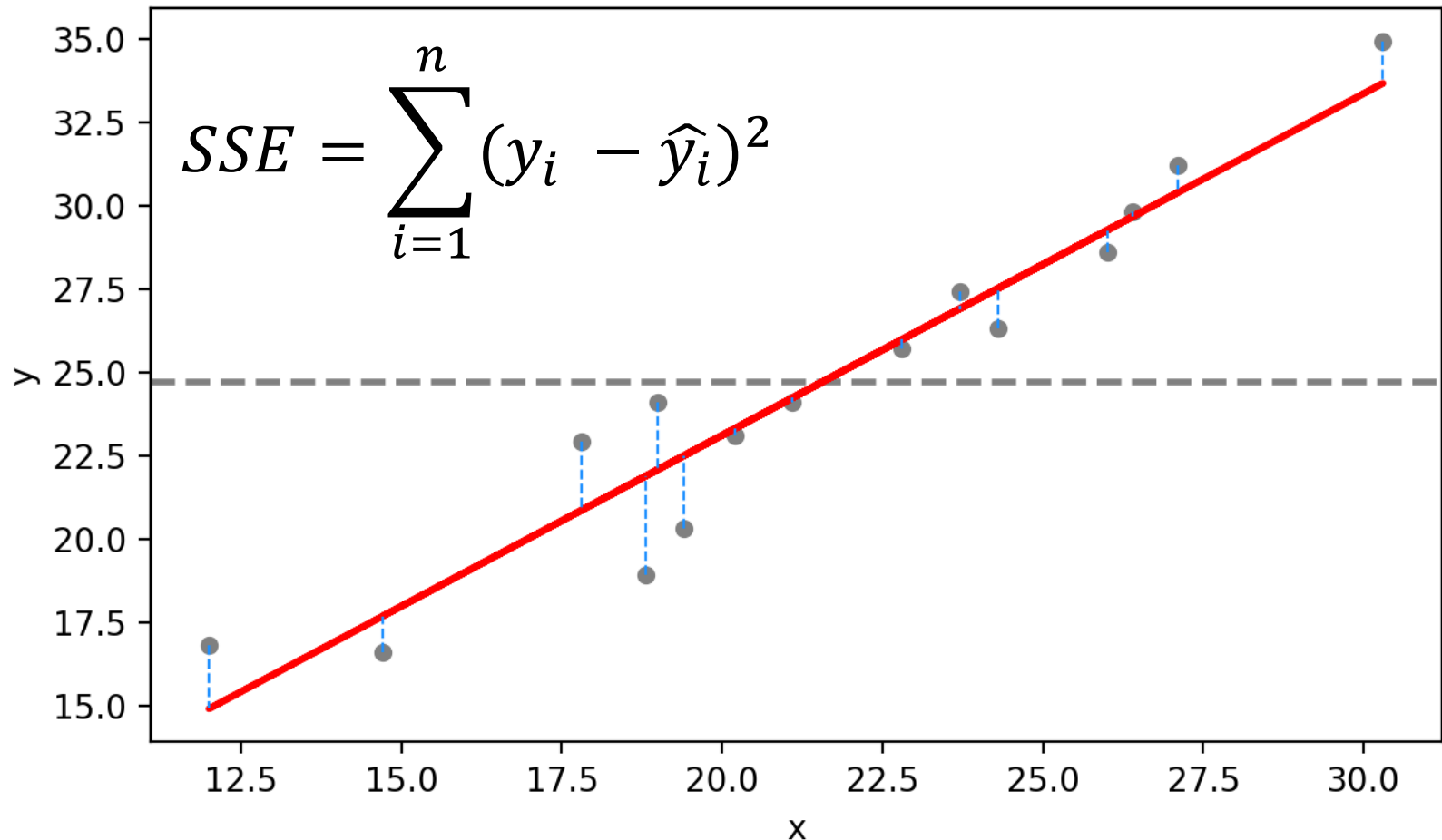


Sum of Squares Total (SST)



Sum of Squares Error (SSE)

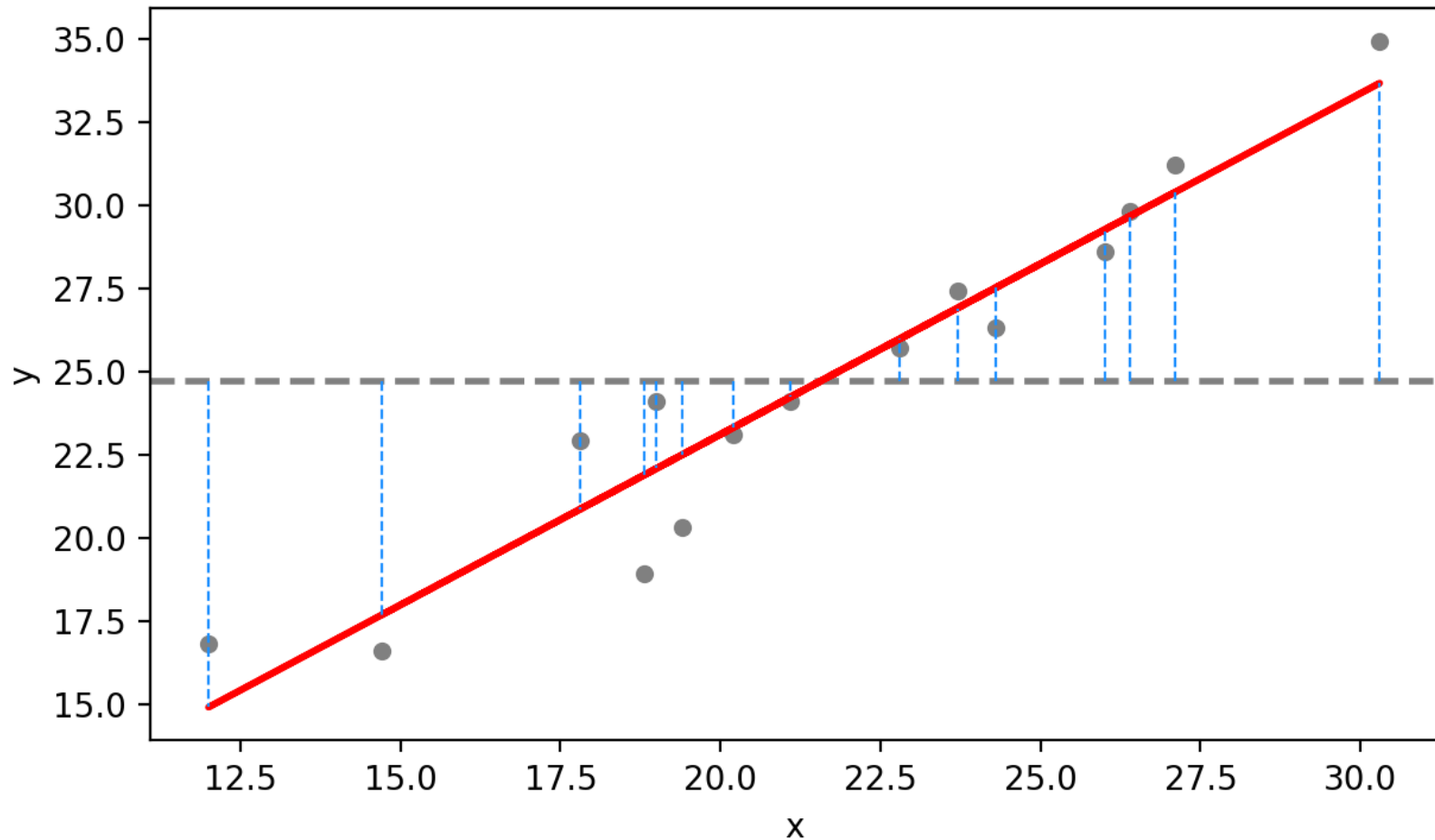
Linear regression line generally fit by minimizing SSE



$\hat{y}_i$ denotes predicted y value using linear regression

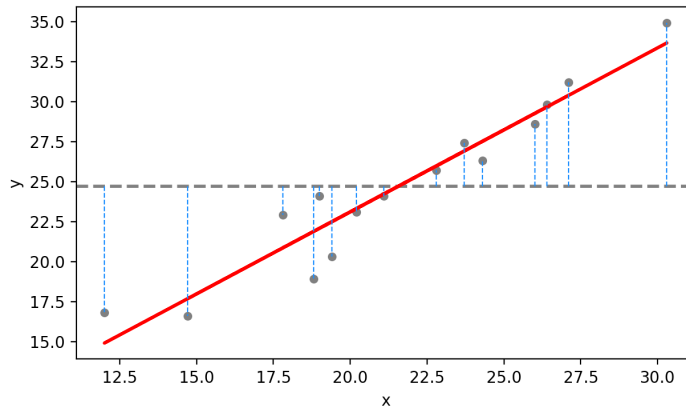
Sum of Squares Regression (SSR)

$$\text{SSR} = \text{SST} - \text{SSE}$$



$$SSR = SST - SSE$$

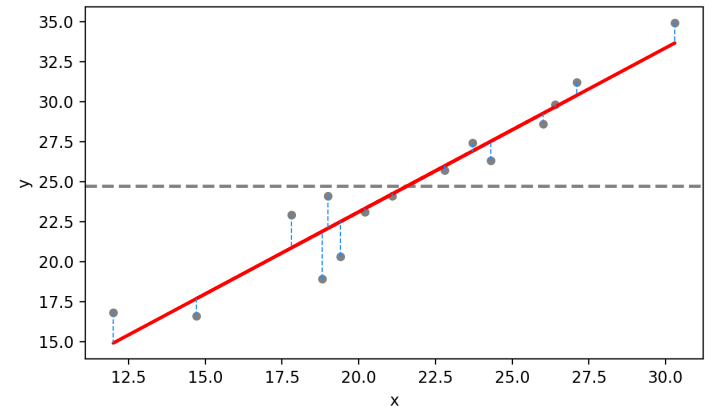
Total Sum of Squares (SST)



Total variance in the model
(relative to no null of no relationship)

■

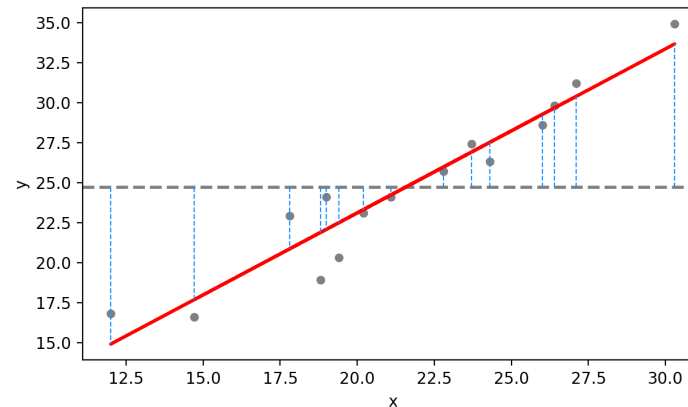
Sum of Squared Errors (SSE)



Variance not explained by the model

||

Sum of Squares for Regression (SSR)



Variance explained by the model

R-squared

- R^2 = sum of squares for regression / total sum of squares
 - Proportion of the total variance explained by the model
 - R^2 of 0.78 says that ~78% of the total variance is explained by the model

```
from scipy.stats import linregress  
lr = linregress(x,y)
```

#Output

```
LinregressResult(slope=np.float64(1.02545287098266),  
intercept=np.float64(2.590896730000747),  
rvalue=np.float64(0.9587532973687167),  
pvalue=np.float64(1.779519609895298e-08),  
stderr=np.float64(0.08431826210956965),  
intercept_stderr=np.float64(1.8624553457473432))
```

R2 = lr.rvalue **2

Multiple Linear Regression

- Rarely is there only one variable which captures most of the variation
- Can extend linear regression to include multiple variables

- Optimizing line of best fit becomes multidimensional

$$\mathbf{y} = \beta_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 \dots + \beta_n \mathbf{x}_n + \epsilon = \boldsymbol{\beta} \mathbf{X} + \epsilon$$


y-intercept


coefficient of
independent variable $\mathbf{x}_n$

Multiple Linear Regression

- We can resolve the intercept and slopes (β) by minimize the residuals (ϵ) and the calculated β is deterministic.
- New Python Package Needed!
 - `scipy` does not have embedded function to conduct multi-linear regression
 - We will use a new package named “`statsmodel`”

```
● @YifanCheng → /workspaces $ source /workspaces/.venv/bin/activate  
○ (ert529) @YifanCheng → /workspaces $ pip install statsmodels scikit-learn
```

Multiple Linear Regressions

```
import statsmodels.formula.api as smf
# Generate Dependent Variable
np.random.seed(42)
n = 300
X1 = np.random.normal(0, 1, n)
X2 = np.random.normal(0, 1, n)
X3 = np.random.normal(0, 1, n)
# True model
Y = 3 + 2*X1 + 0.5*X2 + 1.5*X3 + np.random.normal(0, 1, n)
df = pd.DataFrame({"Y": Y, "X1": X1, "X2": X2, "X3": X3})
X_full = sm.add_constant(df[["X1", "X2", "X3"]])
model_full = sm.OLS(df["Y"], X_full).fit()
```

model_full.summary()

OLS Regression Results						
Dep. Variable:		Y		R-squared:		0.860
Model:		OLS		Adj. R-squared:		0.858
Method:		Least Squares		F-statistic:		604.0
Date:		Wed, 18 Feb 2026		Prob (F-statistic):		8.37e-126
Time:		22:56:37		Log-Likelihood:		-427.08
No. Observations:		300		AIC:		862.2
Df Residuals:		296		BIC:		877.0
Df Model:		3				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
const	3.0984	0.059	52.868	0.000	2.983	3.214
X1	2.0678	0.060	34.737	0.000	1.951	2.185
X2	0.4514	0.061	7.412	0.000	0.332	0.571
X3	1.4918	0.059	25.387	0.000	1.376	1.607
Omnibus:		2.085	Durbin-Watson:		2.115	
Prob(Omnibus):		0.353	Jarque-Bera (JB):		2.051	
Skew:		-0.201	Prob(JB):		0.359	
Kurtosis:		2.955	Cond. No.		1.12	

Analysis of Variance (**ANOVA**)

ANOVA (Analysis of Variance) can be used to analyze how much variability in y is explained by:

- the model as a whole
- each predictor or group of predictors
- changes in model performance when adding/removing predictors

ANOVA answers the following question:

Does the regression model explain more variance than would be expected by random chance?

Analysis of Variance (**ANOVA**)

- Similar to simple linear regression, ANOVA also decompose the variance into explained and unexplained.
- ANOVA partitions the total variability in the response:
$$SST = SSR + SSE$$
- $R^2 = SSR / SST \rightarrow$ overall variance explained by the model
 - **ANOVA F-test** can compare explained variance to unexplained variance in a model.

$$F = \frac{\text{Variance explained by predictor}}{\text{Residual variance}}$$

- Therefore, when $F \gg 1$ and corresponding $P < 0.05$, this regression model or predictor makes significant contribution in explaining the variances.

Three types of ANOVA analyses

Type	Meaning	Intuitive interpretation
Type I	Sequential sums of squares (ss)	“How much does each variable improve the model <i>given that previous variables are already included?</i> ”
Type II	Marginal (unique) sums of squares without interactions	“How much unique variance does each predictor explain after controlling for the other predictors (but ignoring interactions)?”
Type III	Unique sums of squares even with interactions	“What is the effect of this predictor if we account for EVERY other term in the model, including interactions?”

Example

Case study: Predict daily discharge (Q) from precipitation (P), temperature (T), snow water equivalent (SWE)

$$Q \sim P + T + SWE$$

What are the differences between three different types of ANOVA analysis?

Type I (Sequential) ANOVA

- Tests each term (Sum of Squares, **SS**) in the order entered :
 - $SS(X_1)$
 - $SS(X_2 | X_1)$
 - $SS(X_3 | X_1, X_2)$
- Order-dependent; useful for hierarchical or polynomial builds
- NOT appropriate when order is arbitrary and predictors correlate

Type I: If entered as $P \rightarrow T \rightarrow SWE$, early terms may capture shared hydrologic variance

Type II ANOVA

- Tests each main effect controlling for the other main effects
 - $SS(X_1 | X_2, X_3)$
 - $SS(X_2 | X_1, X_3)$
 - $SS(X_3 | X_1, X_2)$
- Ignores interaction terms when assigning SS
- Order-invariant when no interactions and design is reasonably balanced
- Good default for models without interactions

Type II: Tests unique main-effect contributions of P, T, SWE
(no interactions)

Type III ANOVA

- Tests each term controlling for all other terms (including interactions, $Y \sim X_1 + X_2 + X_1X_2$)
 - $SS(X_1 | X_2, X_1X_2)$
 - $SS(X_2 | X_1, X_1X_2)$
 - $SS(X_1X_2 | X_1, X_2)$
- Order-invariant; common in software defaults (SAS/SPSS)
- Preferred with interactions and unbalanced data
- Main effects may be hard to interpret in presence of strong interactions

Type III: With interaction between SWE and air temperature, tests each term controlling for interactions

Example code

```
from statsmodels.stats.anova import anova_lm
# --- Type I (sequential) ANOVA
anova_type1 = anova_lm(model_full, typ=1)
print("Type I ANOVA")
print(anova_type1)
# Type II or Type III
anova_type2 = anova_lm(model_full, typ=2)
print("\nType II ANOVA")
print(anova_type2)
anova_type3 = anova_lm(model_full, typ=3)
print("\nType III ANOVA")
print(anova_type3)
```

What if the independent variables are strongly correlated? Does the following results make sense?

$$Y \sim X_1 + X_2 + X_3 + \epsilon$$

$$X_2 = 0.5X_3$$

Type I ANOVA

	df	sum_sq	mean_sq	F	PR(>F)
X1	1.0	1059.038603	1059.038603	1061.647496	4.532399e-100
X2	1.0	408.099492	408.099492	409.104826	8.509909e-58
X3	1.0	0.520219	0.520219	0.521501	4.707704e-01
Residual	297.0	296.270152	0.997543	NaN	NaN

Type II ANOVA

	sum_sq	df	F	PR(>F)
X1	1111.183750	1.0	1113.921100	1.656958e-102
X2	408.099492	1.0	409.104826	8.509909e-58
X3	408.099492	1.0	409.104826	8.509909e-58
Residual	296.270152	297.0	NaN	NaN

Type III ANOVA

	sum_sq	df	F	PR(>F)
Intercept	2846.665881	1.0	2853.678511	2.387097e-154
X1	1111.183750	1.0	1113.921100	1.656958e-102
X2	408.099492	1.0	409.104826	8.509909e-58
X3	408.099492	1.0	409.104826	8.509909e-58
Residual	296.270152	297.0	NaN	NaN

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